MATRICES, RESCALING AND EIGENVALUES ON ONE PAGE

From PageRank to power iteration

Repeated multiplication finds a persistent direction. Its stretch factor is an eigenvalue.

1 / WHAT PAGERANK HAS ALREADY TAUGHT US

An eigenvector keeps its line under a transformation

For the repaired, damped PageRank matrix G, every column sums to 1. Thus a probability vector keeps its total score without further rescaling. Its limiting vector satisfies

$$G\mathbf{p}^* = \mathbf{p}^*. \qquad \boxed{A\mathbf{v} = \lambda\mathbf{v}, \quad \mathbf{v} \neq \mathbf{0}}$$

The boxed equation is the general definition: $\mathbf{v}$ is an **eigenvector** and λ its **eigenvalue**. Multiplication stretches, shrinks or reverses this vector along its line. In PageRank, $\lambda = 1$ is already known; the iteration discovers the distribution $\mathbf{p}^*$.

2 / TRY A MATRIX WHOSE STRETCH IS UNKNOWN

Multiply, measure the size, rescale

Use

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \mathbf{x}_0 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}.$$

One update of the **power method** is

$$\boxed{\begin{aligned} \mathbf{y}_{n+1} &= A\mathbf{x}_n, \\ s_{n+1} &= \max_i |y_{n+1,i}|, \\ \mathbf{x}_{n+1} &= \mathbf{y}_{n+1}/s_{n+1}. \end{aligned}}$$

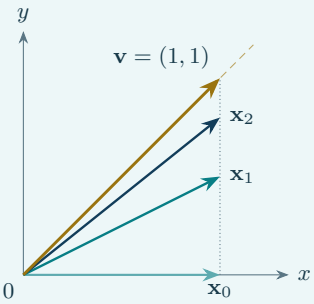
The largest absolute coordinate is now 1. Rescaling preserves the line and controls growth.

First, multiplication sends $(1,0)$ to $(2,1)$. Divide by 2 to get $\mathbf{x}_1 = (1, 1/2)$. Next, $A\mathbf{x}_1 = (5/2, 2)$; divide by $5/2$ to get $\mathbf{x}_2 = (1, 4/5)$.

If $s_{n+1} = 0$, then $A\mathbf{x}_n = \mathbf{0}$: the eigenvalue is 0, and we stop.

3 / SEPARATE DIRECTION FROM SIZE

The arrows approach the line $y = x$



The limiting direction is an eigenvector:

$$A(1,1)^{\mathsf{T}} = 3(1,1)^{\mathsf{T}}.$$

The eigenvalue is 3. Rescaling removes this growth, letting us compare successive directions at the same scale.

4 / FOLLOW THE ARITHMETIC

The removed scale approaches 3

Write $\mathbf{x}_n = (1, t_n)^{\mathsf{T}}$. The update gives

$$t_{n+1} = \frac{1 + 2t_n}{2 + t_n}, \qquad s_{n+1} = 2 + t_n.$$

Here $0 \le t_n \le 1$, so the first output coordinate is the largest.

n	t_n	decimal	s_{n+1}
0	0	0.0000	2.0000
1	1/2	0.5000	2.5000
2	4/5	0.8000	2.8000
3	13/14	0.9286	2.9286
4	40/41	0.9756	2.9756
5	121/122	0.9918	2.9918
∞	1	1.0000	3.0000

A fixed slope satisfies $t(2 + t) = 1 + 2t$, so $t^2 = 1$ and the positive limit is $t = 1$.

5 / WHY DOES THAT DIRECTION WIN?

One component grows faster

Write $\mathbf{v} = (1,1)^{\mathsf{T}}$ and $\mathbf{w} = (1,-1)^{\mathsf{T}}$. Then

$$A\mathbf{v} = 3\mathbf{v}, \qquad A\mathbf{w} = \mathbf{w}.$$

Our starting vector is $\mathbf{x}_0 = \frac{1}{2}\mathbf{v} + \frac{1}{2}\mathbf{w}$. After n multiplications,

$$A^n\mathbf{x}_0 = \tfrac{1}{2}\,3^n\mathbf{v} + \tfrac{1}{2}\mathbf{w} = \frac{3^n}{2}\left(\mathbf{v} + 3^{-n}\mathbf{w}\right).$$

Relative to the growing component, the other is reduced by a factor $1/3$ at each step. Normalising gives the exact formula

$$\boxed{\mathbf{x}_n = \begin{pmatrix} 1 \\ (3^n - 1)/(3^n + 1) \end{pmatrix} \longrightarrow \begin{pmatrix} 1 \\ 1 \end{pmatrix}.}$$

With an eigenvector basis, the same argument works when one eigenvalue has strictly largest magnitude and the start includes its eigenvector component. Starting at $\mathbf{w}$ alone would keep us at $\mathbf{w}$.

6 / RECOVER THE EIGENVALUE AND CHECK THE ANSWER

Does multiplication now behave like a single stretch?

For a real nonzero trial vector $\mathbf{x}$, use the **Rayleigh quotient**

$$\boxed{\widehat{\lambda} = \frac{\mathbf{x}^{\mathsf{T}}A\mathbf{x}}{\mathbf{x}^{\mathsf{T}}\mathbf{x}}}. \qquad \text{If } A\mathbf{x} = \lambda\mathbf{x}, \text{ the quotient is exactly } \lambda.$$

Here $\mathbf{x} = (1,t)^{\mathsf{T}}$ gives $\widehat{\lambda} = 2 + 2t/(1+t^2)$, which approaches 3. Check the **residual** $A\mathbf{x} - \widehat{\lambda}\mathbf{x}$: its entries should be small compared with the largest coordinate of $\mathbf{x}$. Positive rescaling factors recover $|\lambda|$; a negative dominant eigenvalue makes the vectors reverse at each step. The Rayleigh quotient retains the sign.

Notation. ^{T} means transpose; $\mathbf{x}^{\mathsf{T}}\mathbf{y} = \sum_i x_i y_i$ is a dot product.
Scope. This method seeks one dominant eigenpair. A dominant eigenvalue has strictly largest absolute value. Ties in magnitude can cause cycling; near ties slow convergence.
Reading. University of Illinois CS357, *Eigenvalues and Eigenvectors*; Netlib, *Power Method*.

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